

Monosaccharides are linear molecules that cyclize in solution. Shown above are a Fischer and a Haworth projection of a monosaccharide in linear and cyclic form, respectively. What are the configurations (*R* or *S*) of the carbon marked with an asterisk in the linear and cyclic monosaccharide, respectively?

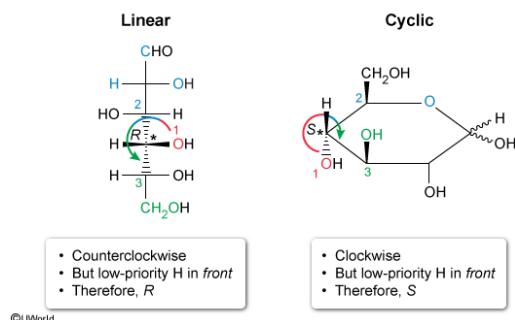
- ☐ A. *R*, *R* [32%]
- ☐ B. *S*, *S* [21%]
- ✓ ☒ C. *R*, *S* [30%]
- ☐ D. *S*, *R* [15%]

Correct

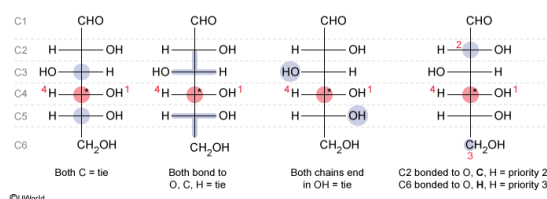
31% answered correctly

Explanation:

Cyclization changes stereochemical designation from *R* to *S*



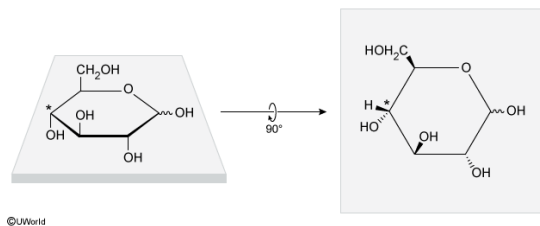
The left image is a **Fischer projection** of a carbohydrate with C4 marked for *R/S* assignment. The –O– and the –H are assigned highest and lowest **priority**, respectively. C3 and C5 tie because both are bonded to an O, a C, and an H, and the highest priority atom (the –O–) on each terminates with an H. Proceeding down the chain, C3 and C5 are bonded to C2 and C6, respectively, resulting in another tie. Finally, comparing C2 and C6, C2 is bonded to an O, a **C**, and an H whereas C6 is bonded to an O and **two Hs**. Therefore, the **C3–C2 substituent takes priority**.



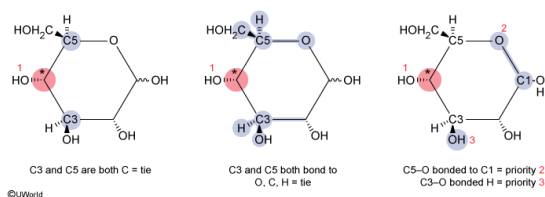
Moving from priority 1 (–OH) to 2 (C3) to 3 (C5) involves a **counterclockwise** movement.

Because the low-priority H is **in front** of the chiral center, C4 in **the linear monosaccharide has an *R* configuration**.

The right image is a Haworth projection, which has the cyclic ring **perpendicular to the viewing plane**. Rotating the molecule so that the cyclic chain is in the plane causes the upward-pointing bonds, including the implied H on C4, to be in front (ie, wedges) and the downward-pointing bonds to be behind (ie, dashes) the plane.



Similar to the linear molecule, the –OH and the –H are assigned highest and lowest priority, respectively, and both C3 and C5 tie. However, in the cyclic molecule the –C3–O– chain terminates with an H but the –C5–O– chain is bonded to C1. Therefore, **–C5–O–C1– takes priority**.



Moving from priority 1 (–OH) to 2 (–C5–O–C1–) to 3 (–C3–O–H) involves a **clockwise** movement. Because the low-priority H is **in front** of the chiral center, C4 in the **cyclic** monosaccharide has an **S configuration**.

(Choices A and B) The linear monosaccharide is *R* because –C5–O–H takes third priority. Upon cyclization, –C5–O–C1– moves to second priority, changing the stereochemistry of C4 to *S*.

(Choice D) In both projections, the low-priority H is *in front of* the chiral center. Therefore, clockwise movement indicates an *S* configuration and counterclockwise movement indicates an *R* configuration.

Educational objective:

R/S absolute configuration for stereochemistry can be determined by assigning priority to a stereocenter's substituents. With the lowest-priority group behind the plane, clockwise movement from priority 1 to 2 to 3 indicates an *R* configuration and counterclockwise movement indicates an *S* configuration. With the lowest-priority group in front, this relationship is reversed.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403624

System:
Functional Groups and Their Reactions